Forecasting the Progression of Humanity on the Kardashev Scale through 2100 A.D. with Machine Learning

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Abstract

The Kardashev scale provides a quantitative framework for relating civilizational development to energy consumption, making long-term energy forecasting central to estimating humanity's future trajectory. However, existing predictions of humanity's progression on the Kardashev scale remain limited and rely on simplified extrapolation of historical trends using linear regression or ARIMA. In this work, we extend previous forecasting efforts to 2100 while developing a more interpretable and informative machine learning framework using gradient-boosted decision tree models to predict future global energy consumption. We incorporate explicit uncertainty estimates and examine the potential effect of additional AI-driven energy demand. Our results suggest that humanity, currently estimated at $K \approx 0.733$, advances only gradually over the coming century, reaching about $K \approx 0.770 \pm 0.002$ in the baseline case and $K \approx 0.777 \pm 0.01$ under a current AI-growth scenario by 2100. The machine-learning framework developed here improves upon previous Kardashev-scale projections by providing a more refined and uncertainty-aware basis for studying long-term trends in human civilizational development.

1. Introduction

Energy provides a useful quantitative lens for examining civilizational development. Over the modern era, human civilization has exhibited sustained growth in total energy consumption, consistent with an increasing population, economic growth, and technological development.[1] In this sense, energy use serves as a natural proxy for the scale and complexity of a civilization. This relationship was formalized by Nikolai Kardashev, who proposed a classification scheme for technological civilizations based on the magnitude of energy they are able to harness and control.[2]

In its canonical form, the Kardashev scale consists of three levels of civilizations. A Type I civilization is capable of harnessing and utilizing all major energy resources available from its home planet, including solar, wind, geothermal, and nuclear energy. A Type II civilization extends this capacity to the scale of its host star, capturing a substantial fraction of stellar output through advanced energy collection. A Type III civilization is able to control energy on the scale of its host galaxy, potentially including energy distributed across billions of stellar systems. Carl Sagan later introduced a continuous formulation of the Kardashev scale through the K-index,

$$K = \frac{\log P - 6}{10} \tag{1}$$

where P is the civilization's power consumption in Watts.[3] Under this formulation, Type I, Type II, and Type III civilizations correspond to power scales of 10^{16}W, 10^{26}W, and 10^{36}W, respectively. In 2024, humanity's total energy consumption was estimated to be 2.1×10^{13}W, placing human civilization at $K = 0.733$ on the Kardashev scale.[4] Humanity remains below Type I, highlighting both the usefulness of the Kardashev framework as a quantitative measure of civilizational development and the large gap between present-day energy use and planetary-scale capability.

Despite extensive work on energy modeling and long-range scenario analysis, quantitative forecasting of humanity's progression on the Kardashev scale remains relatively underdeveloped. Previous work approached this problem using linear regression to extrapolate historical trends in the K-index.[5] While this provides a useful estimate, it oversimplifies the dynamics of civilizational development by treating humanity's progress on the Kardashev scale as a linear function in time, despite historical data suggesting a nonlinear trajectory.

Machine learning (ML) offers a powerful alternative by enabling models to learn complex, nonlinear relationships directly from data. ML approaches are great for tackling problems where a pattern exists, humans cannot pin down the pattern mathematically, and data is available. These methods are well-suited for energy forecasting, where energy consumption depends not only on past trends but also on the interaction of underlying economic, demographic, and environmental drivers. Deep learning (a subset of machine learning focused on artificial neural networks with multiple layers) has been successful in modeling energy consumption. However, this approach has drawbacks: it lacks interpretability and requires a large amount of data to ensure the model does not memorize the training data. In contrast, tree-based ensemble learning models strike a balance among model accuracy, interpretability, the ability to handle noisy data, and the ability to handle limited datasets.

With this motivation, a recent study applied autoregressive integrated moving average (ARIMA) methods to forecast future energy consumption and the Kardashev index through 2060.[6] Although ARIMA is a powerful machine-learning tool, it still forecasts future time series from the past behavior of the series itself. Consequently, it does not fully leverage the strength of machine learning for this problem, since ARIMA cannot incorporate a broader set of real-world factors that shape long-term energy use, including economic development or population changes.

In this work, we build on the effort to use machine learning to forecast humanity's progression by developing a model that incorporates real-world drivers of energy consumption such as economic growth, population dynamics, and geopolitical context. Rather than relying solely on extrapolation of historical data, our approach incorporates external predictors in a gradient boosting model to better capture the nonlinear processes that shape long-term energy consumption. Using these forecasts, we estimate the corresponding evolution of the K-index to 2100, quantify uncertainty in the projected trajectory, and examine how emerging technological trends may alter future energy demand. In particular, we consider the potential influence of increasing energy demand associated with the rapid growth of artificial intelligence. Together, this framework provides a more realistic and interpretable basis for forecasting humanity's future civilizational development.

2. Methods

Our methodology consists of two complementary forecasting stages applied to a set of 42 countries for which sufficiently complete historical and projected input data were available. In the first stage, we train a LightGBM model to forecast the future energy consumption of each country using country-level predictors described in later sections. In the second stage, we train a separate LightGBM model to forecast each country's energy ratio, defined here as its share of global energy consumption, $E_{country}/E_{global}$, using a different set of inputs chosen to capture relative economic scale, demographic scale, and energy intensity. These two components are then combined to reconstruct future global energy-consumption patterns and to estimate humanity's trajectory on the Kardashev scale.The use of two models allows absolute national energy consumption and relative global share to be learned separately, while preserving flexibility in feature choice for each task.

The historical datasets used in this study were assembled from several sources, with population obtained from the United Nations,[7] GDP from the World Bank,[8] global energy consumption from the Energy Institute,[9] and total national energy consumption from World Population Review.[10] Projected GDP values were taken from the Shared Socioeconomic Pathways database maintained by IIASA,[11] and projected population values were obtained from the United Nations.[7]

2.1 Gradient-boosted Decision Tree Models

2.1.1 Model Overview

In this work, we utilize tree-based ensemble learning methods to model the relationship between energy consumption and its underlying drivers. Tree-based ensemble learning methods are widely used for tabular data due to their excellent predictive performance and ability to capture nonlinear feature interactions. Random forest is one such method, combining bagging and random feature selection to reduce variance and overfitting [12]. In contrast, gradient boosting machines construct decision trees sequentially, with each new tree trained to correct errors made by the previous ensemble, allowing the model to iteratively refine its predictions and capture complex patterns[13]. Popular implementations of tree-based gradient boosting machines include LightGBM[14], XGBoost[15], and CatBoost[16].

Here, we use LightGBM because it provides fast training while maintaining strong predictive accuracy. LightGBM differs from gradient boosting machines by avoiding heavy computation by not processing all data. LightGBM uses Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). GOSS prioritizes sampling instances with larger gradients (or larger feedback), and EFB combines features together, allowing for faster training. LightGBM uses histogram-based split finding during tree construction by discretizing continuous features into bins. LightGBM also uses leaf-wise growth, which results in more splits and deeper trees in regions with the highest error.

2.2.1 Model I (LightGBM Energy Consumption per Country)

We train a LightGBM model to predict the energy consumption of a set of 42 countries that play major roles in the global economy and development. The model was trained using selected features from ecological footprint, GDP, population, biocapacity, year, urban population, country code, temperature, and precipitation. The final feature set was restricted to GDP and population, as these variables exhibited high SHAP importance in the trained model (Fig. I) and were among the few features for which sufficiently consistent projection data were available over the full forecasting period.

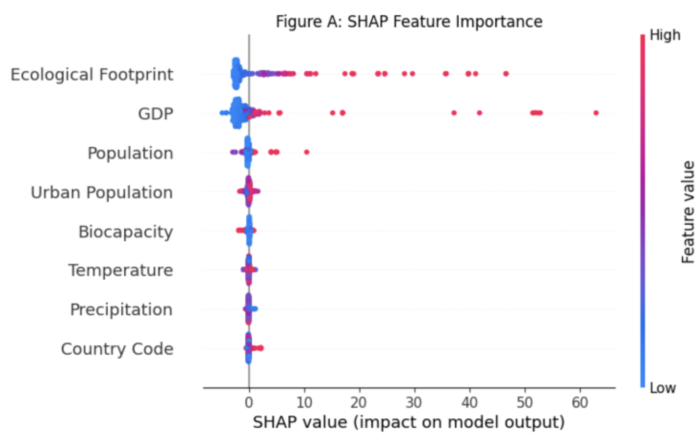


Figure 1. *Input features ordered by mean SHAP value. The three most important features influencing the model output are ecological footprint, GDP, and population.*

For model training, the dataset was split into 80% training set and a 20% validation set. The LightGBM model was specifically trained using gradient boosting decision trees (boosting_type="gbdt") with default hyperparameters and 100 boosting iterations. Model performance was evaluated using L2 loss (mean squared error), computed on both the training and validation sets during each boosting iteration. We recorded training and validation loss curves throughout training to assess generalization performance.

2.2.2 Model II (LightGBM ratio)

To model the evolution of a country's share of global energy consumption, we selected predictors representing relative economic scale, relative population, and energy intensity: $GDP_{country}/GDP_{global}$, $Population_{country}/Population_{global}$, and $E_{country}/GDP_{country}$. This choice is supported by historical evidence that long-run energy use is strongly shaped by the scale of economic activity and population, as well as by the amount of energy required to generate output.[17] Historical energy analyses show that growth in economic activity and population are major drivers of energy demand,[18] making relative GDP and relative population natural predictors of each country's energy consumption share. Energy intensity is also directly relevant because it determines how strongly a given level of economic output translates into energy use. Changes in energy intensity have been linked to structural change, technological progress, and shifts in production conditions, and can also reflect the effects of energy and climate policy to the extent that such policies alter efficiency, fuel choice, and patterns of energy use.[19] Policy and price-driven efficiency improvements have historically played an important role in restraining aggregate energy demand growth, even when they have not fully offset the effects of economic growth.[20]

To train the second model (LightGBM), the features were engineered at the country level using the same 42 countries as in Model I and covering the period from 1970 to 2019. The input features were relative economic scale, relative population, and energy intensity, so X = ($GDP_{country}/GDP_{global}$, $Population_{country}/Population_{global}$, $E_{country}/GDP_{country}$), and the target variable y = $E_{country}/E_{global}$. The three selected features, X, were then computed for each observation from 1970 to 2100. The resulting dataset was divided into an 80% training set and a 20% validation set using a fixed random split. We trained a LightGBM regressor using gradient-boosted decision trees with hyperparameters tuned via a grid search, testing combinations of tree depth, learning rate, and number of estimators to improve validation performance without overfitting to the training set. The final model used 300 boosting iterations, a learning rate of 0.05, and a maximum tree depth of 6. Model performance was evaluated using the validation R^2 score, which remained high (0.997), indicating that the selected predictors explain most of the variation in the ratio target across countries and years. We also plotted training and validation curves to ensure the model is not overfitting.

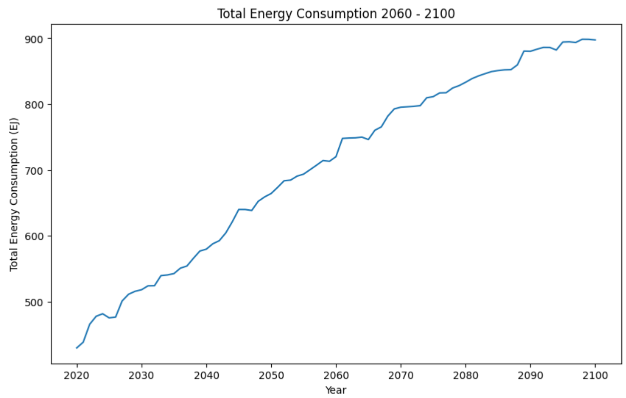
To generate forecasts from 2020 to 2100, projected GDP and population values were combined with the energy-consumption predictions from Model I to compute the same three input features for each country, since we are using supervised machine learning. The trained LightGBM ratio model was then applied to predict each country's future share of global energy consumption, and these country-level ratios were summed by year to estimate the aggregate E_{42}/E_{global}, where $E_{42} = \sum_{42\ countries} E_{country}$. SHAP analysis was used after training to confirm that all three variables contributed meaningfully to the model output. The resulting ratio forecast predicted for each country by Model II is then combined with Model I energy predictions to reconstruct projected global energy consumption.

2.2.2 AI Energy Demand Forecast

To estimate the potential impact of artificial intelligence on future energy demand, we used the IEA annual projections for global data center energy consumption over 2020-2030,[21] since recent data-center related demand is driven in large part by the increasing deployment of AI-oriented accelerated servers. Because this dataset is very small, machine-learning methods are not appropriate for this component of the study; we instead apply least-squares fitting to several candidate functional forms and select the model that best represents the projected trend. Among the fits considered, a quadratic model provided the strongest fit and was therefore adopted as the central estimate for extrapolation beyond 2030. The second and third best fits, given by the logistic and Gompertz functions, were used to define lower and upper bound uncertainty ranges. These alternative fits are not treated simply as a fitting error, but as representing the high uncertainty in future AI energy demand evolution. In particular, uncertainty in the continued acceleration or earlier saturation in energy demands and in the extent of hardware and software efficiency improvements can lead to substantially different long-term growth shapes. The extrapolated AI-related energy demand was then added to the broader global energy forecast to estimate its effect on humanity's projected trajectory on the Kardashev scale.

Results

3.1 Total Energy Consumption Prediction 2100



Global energy consumption, using the trained model (model I), is projected from 2020 to 2100 based on projections of GDP and population through 2100.

Figure 2. *Total Energy Consumption vs Year. Total energy consumption increases monotonically from 430 EJ in 2020 to about 900EJ in 2100, roughly doubling over 80 years.*

In Figure 2, it can be seen that projected global energy consumption for the selected countries is monotonically increasing, with small fluctuations. Below is Figure 3, showing that the model does not overfit and generalizes well.

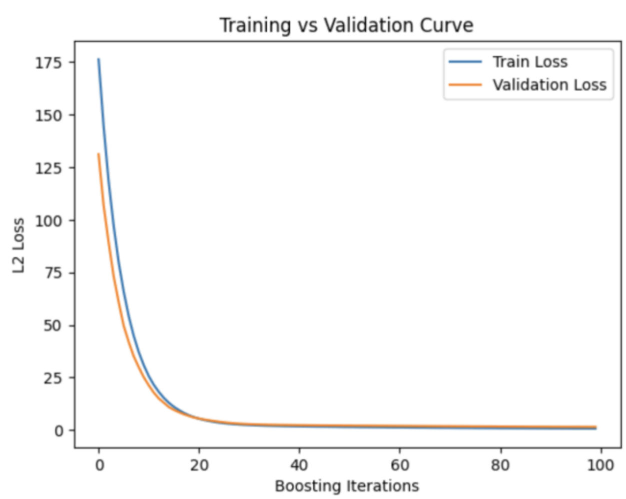


Figure 3. *Training and validation loss during training of the model I over 100 boosting iterations. L2 loss is used to measure accuracy. The training and validation losses decrease sharply until around 20 boosting iterations, then plateau. Since the validation loss does not increase while the training loss decreases (and the two curves track closely), there is no significant indication that the model is memorizing the data.*

3.2 Feature Importance Based on Model II

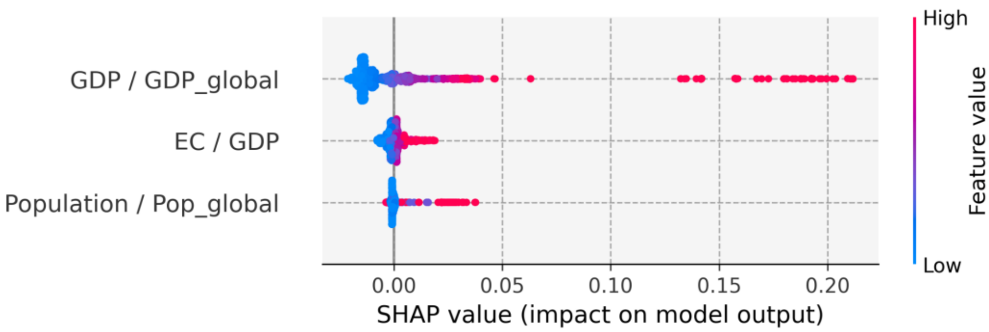


Figure 4. *Input features representing relative economic scale, energy intensity, and relative population are ordered by mean SHAP value. The most important feature is relative economic scale $GDP_{country}/GDP_{global}$ (listed at the top), while relative population contributes the least but still influences model output.*

SHAP analysis of the Model || ratio forecast shows that the selected variables capture most of the variation in the historical E_{42}/E_{global} ratio while remaining interpretable. Figure 4 shows that relative economic scale ($GDP_{country}/GDP_{global}$) has the largest mean SHAP value, indicating that it contributes the most to predicting a country's share of global energy consumption. This is consistent with the fact that countries with larger shares of global economic output generally account for larger shares of total energy use, since production, infrastructure, and industrial activity are closely linked to energy demand. Energy intensity ($E_{country}/GDP_{country}$) also has a clear contribution, showing that countries requiring more energy per unit of economic output tend to have a larger energy share even when total GDP is similar. Relative population $Population_{country}/Population_{global}$ contributes less strongly than the other two features, but still captures meaningful variation by reflecting the demographic scale underlying long-run energy demand. The high validation R^2 obtained by the ratio model further supports that these three features together explain most of the historical contributors to this ratio, indicating that the model generalizes well and that the selected predictors are sufficient for long-range forecasting and relative global energy share.

3.3 Energy Ratio Prediction & Global Energy Prediction

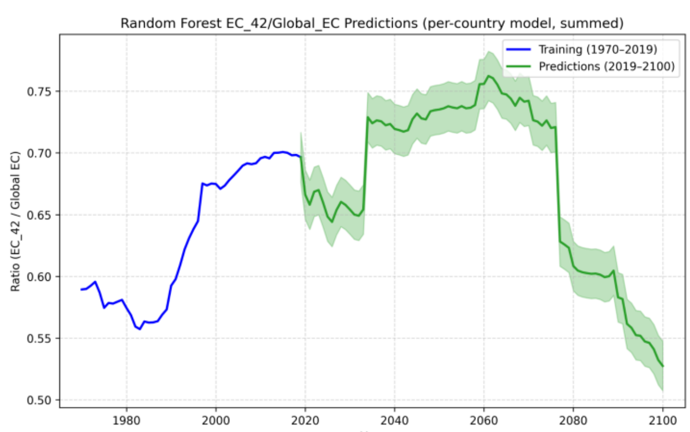


Figure 5. *Historical and forecasted ratio of energy consumption for the 42 modeled countries relative to global energy consumption, E_{42}/E_{global}. The blue curve shows historical values from 1970–2019, while the green curve shows Model II predictions from 2019–2100. The shaded region represents uncertainty added to reflect historical deviations not captured by the selected predictors.*

From Figure 5, we can see the summed country-level ratio forecast produced by Model II, in which the predicted country shares are aggregated to estimate the future trajectory of E_{42}/E_{global}. The historical training curve rises from about 0.59 in 1970 to nearly 0.70 by 2019, indicating that the 42 countries in the dataset account for a large fraction of global energy consumption over the historical period. In the forecast region, the ratio initially remains relatively stable through the mid-century range, fluctuating around 0.72–0.75 before declining after approximately 2075, suggesting that countries outside the selected dataset account for an increasing share of future global energy growth. The step-like variations visible near 2035 and 2075 arise from discontinuities in the projected economic and demographic input data and are carried through the ratio model.

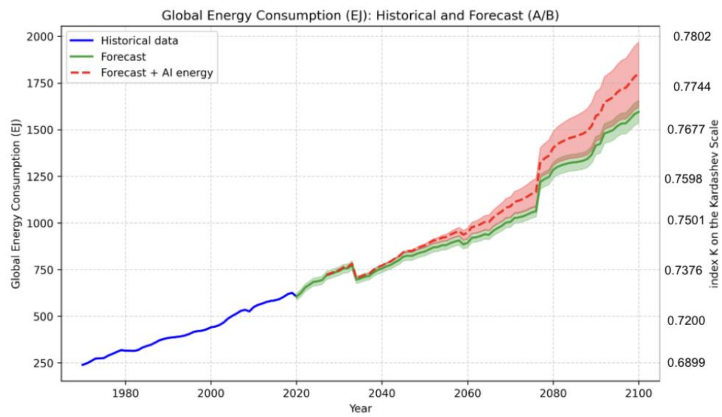


Figure 6. *Projected trajectory of global energy consumption until 2100 using Model I and Model II outputs. The plot compares a baseline forecast (green) against energy demand driven by an AI forecast (red). The red-shaded region reflects the high uncertainty in the impact of AI. The shaded bands indicate uncertainty ranges, and the right axis shows the corresponding Kardashev index.*

	Projected values		
		K	
Year	Energy consumption (EJ)	Value	Growth rate (past 5 years)
2025	687.04	0.73382	7.63×10^{-3}
2030	742.26	0.73718	4.58×10^{-3}
2035	702.38	0.73478	-3.25×10^{-3}
2040	755.23	0.73793	4.29×10^{-3}
2045	819.57	0.74148	4.81×10^{-3}
2050	846.90	0.74290	1.92×10^{-3}
2055	881.09	0.74462	2.31×10^{-3}
2060	893.10	0.74521	7.89×10^{-3}
2065	935.87	0.74724	2.73×10^{-3}
2070	1004.42	0.75031	4.11×10^{-3}
2075	1056.03	0.75249	2.90×10^{-3}
2080	1283.74	0.76097	1.13×10^{-3}
2085	1324.44	0.76232	1.78×10^{-3}
2090	1415.19	0.76520	3.78×10^{-3}
2095	1518.02	0.76828	4.02×10^{-3}
2100	1594.94	0.77039	2.76×10^{-3}

Table 1. *Final forecasting results. The predicted values for energy consumption, civilization development index K, as well as its growth rate over each 5-year period until 2100.*

Using the predicted ratio together with the country-level energy forecast from Model I produces the final global energy-consumption trajectory shown in Figure 6, and the corresponding Kardashev value predictions are displayed in Table I.

To account for abrupt deviations from the long-term trend of E_{42}/E_{global} that are not captured by the selected predictors, we include an additive constant uncertainty term of 0.4 in the energy-ratio forecast. This value is motivated empirically by the magnitude of historical fluctuations around the broader trend, particularly during periods of sudden transition seen between the years 1970 and 2000. These deviations likely reflect external shocks not explicitly represented in the model, such as geopolitical disruptions, rapid changes in energy markets, or other large-scale structural events. Incorporating this constant uncertainty term allows the projected ratio to reflect the degree of variability historically associated with such periods of instability, thereby providing a more realistic estimate of future trajectories.

The global forecast, shown in Figure 6 continues increasing throughout the century, rising from about 620 EJ near 2020 to approximately 1595 EJ by 2100 in the baseline case (without accounting for AI). Although growth remains steady, the slope is insufficient to approach the Type I civilization threshold, and the corresponding Kardashev index reaches only $K \approx 0.770 \pm 0.002$ by 2100. Small variations in slope appear where the ratio forecast shifts, but the long-run trend remains monotonic overall. The AI-adjusted scenario, shown by the dashed red curve, increases the upper range of projected energy demand while preserving the same general trajectory.

3.4 AI Energy Consumption

This AI scenario assesses the potential contribution of the continued rapid advancement and widespread adoption of AI technologies. The trajectory of AI development is highly uncertain, with no consistent historical pattern that can reliably predict the scale, use, and efficiency of future AI innovations. Under the best-fit quadratic model, global energy consumption increases more rapidly after 2030, and a divergence between the baseline prediction and the AI-trajectory is observed. By 2100, it can be seen that the increase in energy consumption due to AI is on the order of 0.005-0.01 on the Kardashev scale. Specifically, for the 2100 global energy consumption forecast, the Kardashev value on the AI growth track is ~0.776 ± 0.01 (or 1799.860 EJ/year ± 174.953 EJ/year), and the Kardashev value on the non-AI track is ~0.770 ± 0.002 (or 1594.935 EJ/year ± 60.476 EJ/year). In a higher-growth scenario, the increase in energy consumption is on the order of 0.01-0.02, nearing 0.78 on the Kardashev scale; in a lower-growth scenario, it follows the baseline quite closely. If AI becomes very energy efficient (through advances in GPU and TPU efficiency, algorithm optimization, or datacenter design), the contribution of AI to the global energy demand may resemble the lower estimate or the baseline energy consumption forecast. Additionally, if the AI bubble bursts and AI usage/development slows, the energy demand associated with AI would peak and then decline, limiting AI's impact on global energy consumption.

Discussion

The results suggest that under continued growth in global energy demand, humanity is projected to remain well below Type I civilization by 2100. Using a forecasting framework that incorporates projected demographic, economic, and efficiency-related drivers, we estimate a 2100 Kardashev index of approximately $K = 0.770 \pm 0.002$ in the baseline case and $K = 0.776 \pm 0.01$ when additional AI-related energy demand is included. These values indicate measurable continued civilizational advancement and support a picture of gradual rather than abrupt Kardashev-scale progression.

A key result of this study is that modeling the energy ratio using projected real-world drivers yields a more informative forecast than direct extrapolation of the ratio as a time series. Earlier approaches based on linear regression or ARIMA provide useful benchmarks, but they treat future development primarily as a continuation of past trends. By contrast, our model links changes in E_{42}/E_{global} to relative population, relative GDP, and energy intensity, allowing the forecast to respond to structural features of the real world. This improves both interpretability and predictive flexibility, and better captures finer variation in the projected ratio. In addition, the inclusion of an explicit uncertainty range makes it possible to evaluate not only the most likely trajectory predicted by the model, but also the extent of variation that may arise from historically observed departures from the trends and the model's predictions.

The AI-demand analysis further shows that emerging digital infrastructure may become a significant contributor to long-term energy consumption, but does not substantially alter the Kardashev conclusion. Although the AI-growth track produces a higher projected K index than the base case, the difference remains modest relative to the large gap between present-day civilization and the Type I threshold.

At the same time, the forecast remains subject to high uncertainty. Both models of the forecasting framework are trained on data from only 42 countries, and therefore do not capture future changes in countries outside this sample. This is especially consequential for the ratio model, since as countries outside the dataset continue to develop, their changing contribution to global energy consumption may introduce variation that the ML models cannot account for. In addition, both models rely on projected inputs such GDP and population, which are themselves uncertain over long time scales. This uncertainty is not solely a limitation of the present framework, but a reflection of the forecasting problem itself. Long-horizon predictions of global energy use are inherently uncertain because energy consumption depends on coupled demographic, economic, technological, and geopolitical developments, so the results are most appropriately interpreted as a range of plausible trajectories rather than a single outcome.

Conclusions

In this work, we developed a machine learning framework to forecast humanity's progression on the Kardashev scale through 2100 by combining two complementary LightGBM models for country-level energy consumption and global energy share. The main methodological improvement over prior work is replacing the ARIMA-based ratio forecast by engineering new features and training a new model that uses relative GDP, population, and energy intensity to predict E_{42}/E_{global}. This allows the forecast to respond to underlying economic and demographic drivers rather than relying only on past trends, improving both interpretability and long-term reliability. Using this framework, our results suggest that while global energy consumption continues to grow steadily, humanity is unlikely to reach Type I civilization by 2100, remaining at approximately $K \approx 0.77$ even under an AI-driven growth scenario.

One area of future work is scenario-based forecasting to better capture the wide range of possible technological advances and policy-driven influences on global energy consumption. In addition to the baseline and AI-case considered, other scenarios could include: a fusion energy development scenario (where fusion energy development increases the amount of power humanity consumes) and a sustainability scenario (where energy consumption becomes more efficient). Additionally, the current analysis is limited to 42 countries due to data availability constraints, introducing uncertainty when modeling the global energy ratio. In future work, it would be important to expand the dataset to include more developing countries with developing economies in Africa, Asia, and South America. This extension would better capture shifts in the global energy demand driven by developing countries. Another direction for future work is to develop a probabilistic uncertainty model rather than using a fixed uncertainty term, allowing uncertainty to evolve dynamically with changes in economic and geopolitical conditions. In addition, future models could replace the current tree-based approach with more advanced machine learning methods that better capture large-scale economic dynamics, such as models specifically designed for long-horizon macroeconomic forecasting but still work with limited data.

Appendix

Relation to previous ARIMA paper

Previous work has used machine learning to predict human civilization's energy consumption in 2060 on the Kardashev scale. The work collected data on ecological footprint, GDP, population, biocapacity, year, urban population, country code, temperature, and precipitation from 1960 to 2020 for a select set of 42 countries. A select number of countries were studied due to limited data availability. The countries for which data were collected play major roles in the global economy and development. A random forest model is trained on GDP, population, biocapacity, year, urban population, country code, temperature, and precipitation from 1960 to 2020 to predict the ecological footprint for a given country for a given year. Projections of the input features from 2020 - 2060 are used as input to the random forest model to predict the sum of energy consumption for a select set of 42 countries ($E_{42\_Random\_Forest\_Prediction}$). Then, an Autoregressive Integrated Moving Average Model (ARIMA) is trained on the historical ratio of the energy consumption of a select set of 42 countries to the global energy consumption per year (E_{global}). The ARIMA model is then used to forecast the E_{42}/E_{global} ratio for 2020 - 2060. Then, the Kardashev value is found using the calculated global energy consumption from the formula: $E_{42\_Random\_Forest\_Prediction}$ / (E_{42}/E_{global}). The random forest does not directly compute global energy consumption due to a lack of data, specifically projection data for all countries.[21]

ARIMA is a general class of models used for forecasting time series data that is generally good for time-series modeling and capturing trends and seasonal variations in the data. The ARIMA takes as input parameters p, d, and q. The parameter p is the number of autoregressive terms or lags in the stationary time series forecasting equation, where stationary means that the statistical properties of the equation are constant. The parameter d is the number of nonseasonal differences needed for the stationary equation. The parameter q is the number of lagged forecast errors in the prediction equation.[22]

Although ARIMA is one of the most widely used approaches to time series forecasting, there are several shortcomings when ARIMA is used to forecast the ratio of country-level energy consumption to global energy consumption in previous literature. In particular, ARIMA forecasts poorly when trained on a small dataset, and the historical ratio dataset is small, containing only 60 datapoints from 1960 to 2020. With such a small dataset, ARIMA is better suited for short-range predictions rather than long-term forecasting, making projections until 2060 unreliable.[23] ARIMA additionally struggles to capture turning points in the data as ARIMA fails to accurately predict sudden changes. This is due to the fact that ARIMA relies on historical time series patterns of a ratio rather than taking into account real-time real-world factors such as geopolitics, economics, and population dynamics.[24] These factors are dynamic and are driven by real events that can not be captured well using historical patterns alone. Furthermore, the ratio itself is impacted heavily on energy consumption from countries outside the selected dataset, reducing the reliability of the ARIMA forecasts.

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